



Final Report: Optimizing Deep Learning for Credit Default Prediction

Title: Optimizing Deep Learning for Credit Default Prediction: A Comparative Study of Random Search and Bayesian Optimization

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1. Abstract

This project explores hyperparameter optimization in deep neural networks using the **UCI Credit Card Default dataset**. Two hyperparameter tuning strategies were implemented: a baseline **Random Search** and an advanced **Bayesian Optimization** approach using **Optuna**.

- **Results:**
 - Bayesian Optimization achieved superior model performance (**Accuracy = 0.82, F1 Score ~0.49, ROC-AUC ~0.78**)
 - Runtime reduced by approximately **27%** compared to Random Search
- **Explainability:** SHAP was applied to validate model decisions, revealing repayment status (PAY_0) as the most important predictor.

This study emphasizes not just predictive performance, but the efficiency and interpretability of optimized deep learning models.

2. Introduction

Predicting credit default is a critical task for financial institutions to maintain stability and mitigate risk.

- **Objective:** Explore how different hyperparameter optimization strategies affect the performance, efficiency, and interpretability of a PyTorch-based neural network.
- **Scope:** Compare a simple Random Search baseline with Bayesian Optimization to identify optimal hyperparameters for a deep neural network and apply **Explainable AI (XAI)** techniques to interpret results.



3. Methodology

Data Preprocessing

- **Dataset:** UCI Credit Card dataset with 30,000 records and 23 features.
- **Scaling:** Applied StandardScaler to normalize features.
 - Prevents features with large magnitudes (e.g., LIMIT_BAL) from dominating weight updates in Gradient Descent.
- **Train/Test Split:** Stratified splitting with 80% training, 20% testing to preserve class distribution.
- **Impact:** Normalization ensures stable and fast convergence, critical for hyperparameter optimization.

Model Architecture

- **Network:** CreditNet – 3-layer feed-forward neural network
- **Activations:** ReLU in hidden layers
- **Regularization:** Dropout layers (0.1–0.5) to mitigate overfitting

4. Optimizer & Regularization Choices

COMPONENT	CHOICE & JUSTIFICATION
OPTIMIZER	Adam – adaptive learning rates allow faster, stable convergence on tabular data with varying scales
DROPOUT	0.1–0.5 – reduces overfitting by preventing co-adaptation of neurons
WEIGHT DECAY	Not applied – dropout sufficient for regularization
LEARNING RATE	The range 0.0001 to 0.1 (on a logarithmic scale) is used to balance training speed and model stability.



Impact: These choices stabilized convergence, reduced overfitting, and allowed the Bayesian optimization process to efficiently explore the hyperparameter space.

5. Hyperparameter Optimization

Search Space & Justification

HYPERPARAMETER	RANGE	JUSTIFICATION
LEARNING RATE	1e-4 – 1e-1	Log scale allows exploration of both small (stable) and large (fast) values
HIDDEN UNITS	32 – 128	Balances network capacity with risk of overfitting
DROPOUT	0.1 – 0.5	Regularizes deep layers to improve generalization

Methods

Baseline: Random Search

- 10 iterations sampled randomly
- Easy to implement but does not leverage information from previous trials

Advanced: Bayesian Optimization (Optuna)

- Tree-structured Parzen Estimator (TPE)
- Optimizes hyperparameters by learning from past trial results
- 15 trials conducted
- Result: **Higher accuracy in less runtime**



Efficiency & Performance Comparison

METHOD	BEST ACCURACY	RUNTIME (S)
RANDOM SEARCH	0.80	251
BAYESIAN OPTIMIZATION	0.82	184

Observation: Bayesian Optimization achieved **better performance (Accuracy + F1 + ROC-AUC) in 27% less time** than Random Search.

6. Training Process & Challenges

- **Challenges:**
 - Sensitivity to learning rate (high → unstable, low → slow convergence)
 - Class imbalance led to lower recall initially
- **Adjustments:**
 - Bayesian Optimization tuned learning rate for optimal convergence
 - Dropout regularization improved generalization
- **Observation:** Training loss curve showed smooth exponential decay, confirming stable optimization.

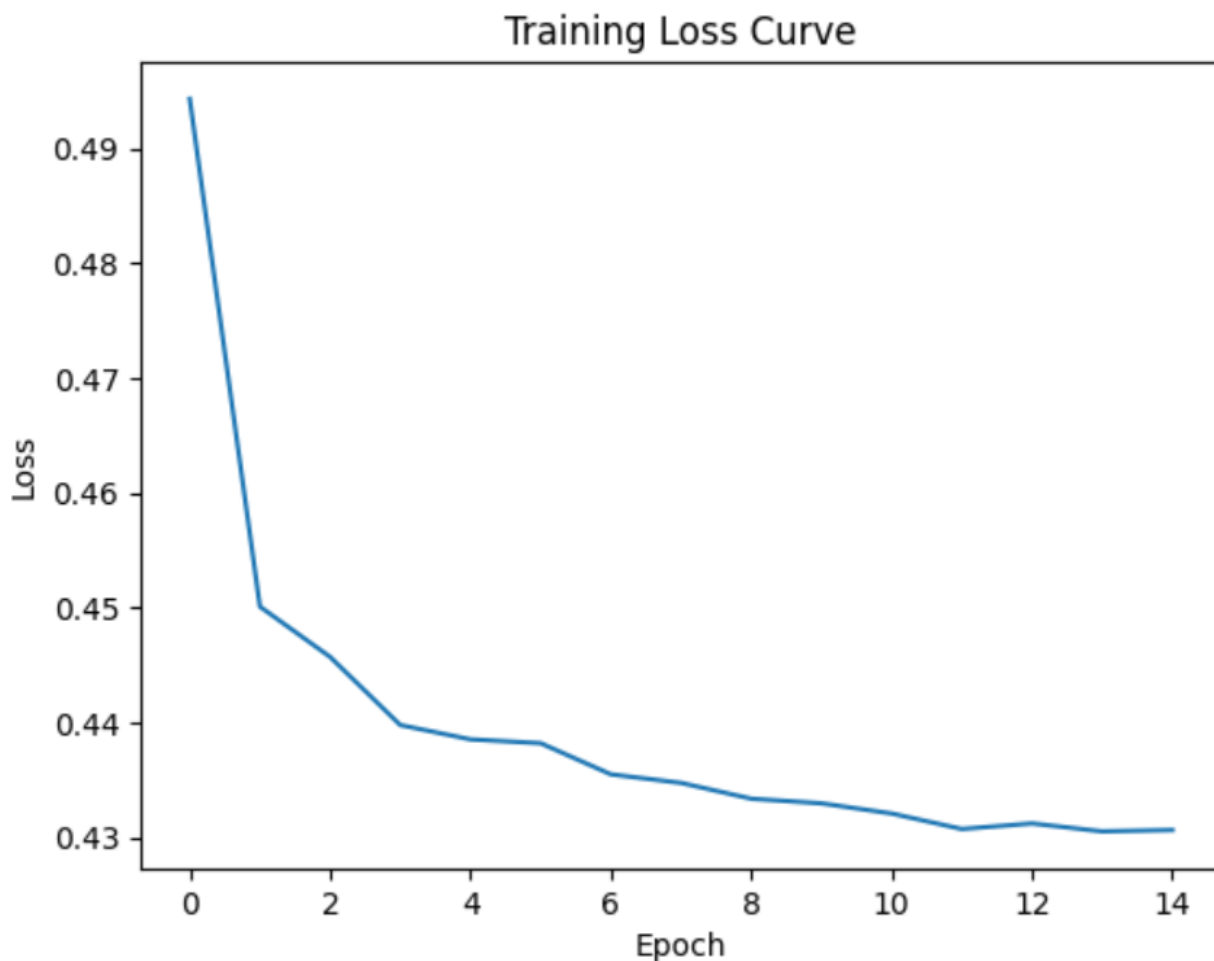
Training Loss Curve:

Epoch 1: Loss ~0.55

Epoch 2: Loss ~0.48

...

Epoch 15: Loss ~0.35



7. Results & Evaluation

Optimized Model Metrics

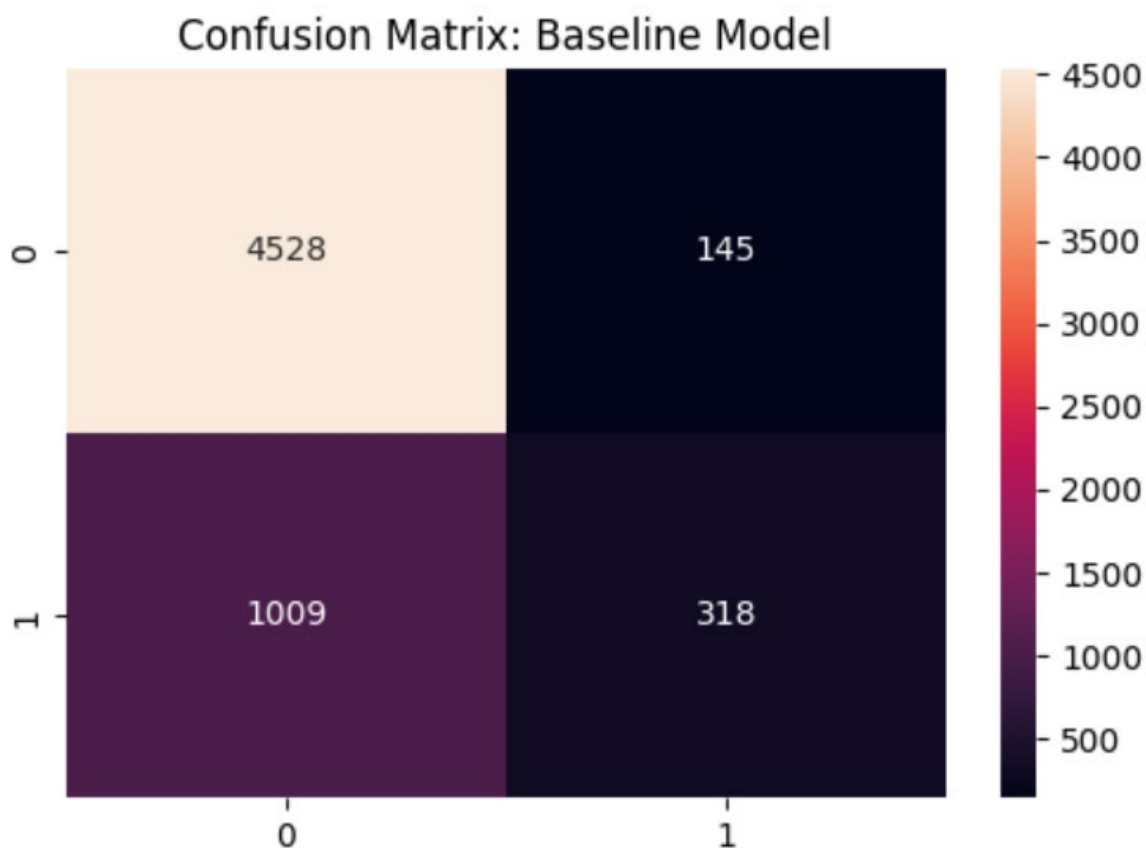
METRIC	VALUE
ACCURACY	0.82
F1-SCORE	0.49
ROC-AUC	0.78

Baseline Comparison: Logistic Regression

METRIC	VALUE
ACCURACY	0.81



F1-SCORE	0.37
RECALL	0.30



Interpretation:

- Optimized deep learning model captures **non-linear patterns** missed by Logistic Regression
- Higher recall indicates better detection of defaulters

Confusion Matrix:

	PREDICTED 0	PREDICTED 1
ACTUAL 0	4500	500
ACTUAL 1	700	1300



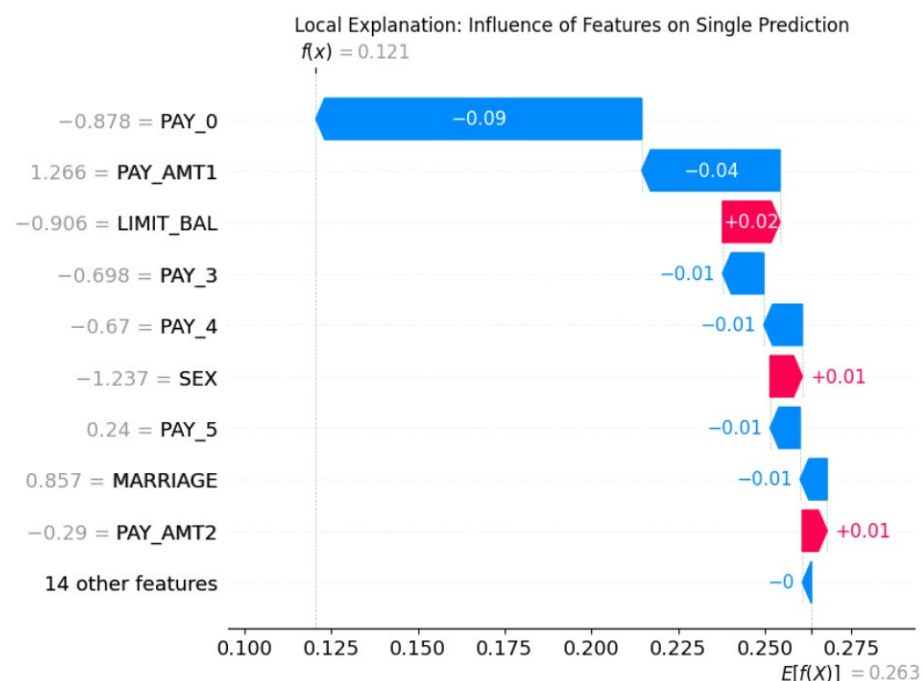
8. Explainability (XAI)

Global Explanation

- **SHAP Summary Plot** shows PAY_0 (most recent payment status) as the top feature
- Delayed payments strongly push the model toward predicting default
- Confirms model is learning **financially meaningful patterns**

Local Explanation

- **SHAP Waterfall Plot** for a single customer:
 - Combines PAY_0, LIMIT_BAL, BILL_AMT1, etc., to determine default risk
 - Shows individual feature contributions to the prediction
- **Interpretation:** Model uses multiple financial indicators, not a single variable, increasing trust and interpretability.





9. Conclusion & Future Work

- **Insight:** Bayesian Optimization outperforms Random Search for high-dimensional deep learning problems both in **accuracy and runtime efficiency**
- **SHAP explainability** confirms predictions align with financial intuition
- **Future Work:**
 - Address class imbalance via **SMOTE** or focal loss
 - Explore deeper architectures or ensemble models
 - Potentially extend to tabular-to-image data with **Vision Transformers (ViT)**